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Transformer is All You Need

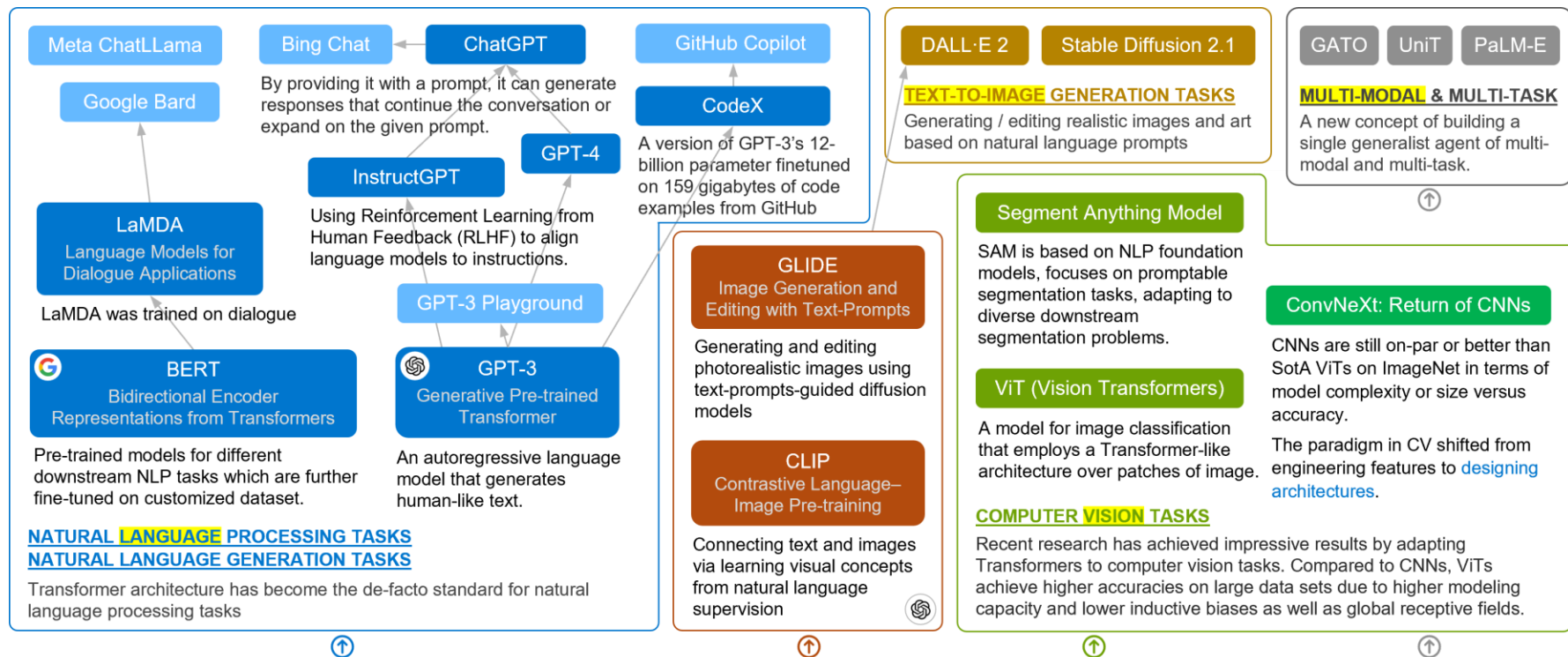
2

Large Language Models

3

Generative AI Techniques

Transformers Take Over Artificial Intelligence



Transformers Large-scale Transformer models triggered breakthrough in AI-Generated Content

A Transformer is a type of neural network architecture based on the concept of self-attention, which allows the model to focus on specific parts of the input sentence to better understand the context of the sentence. Transformer models have been used to achieve state-of-the-art results in many NLP tasks, such as machine translation, text summarization, and question answering.

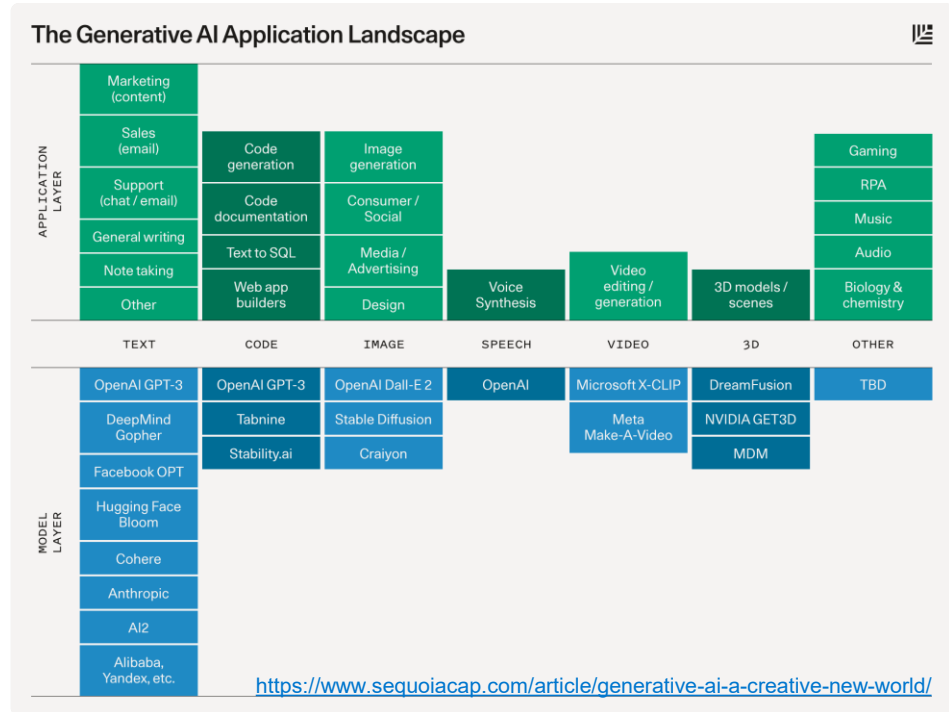
Generative AI Techniques

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Apr 2023

Introduction to Generative AI

- The term “generative AI” is used to describe AI systems that can **create new information from scratch**, as opposed to merely evaluating or acting on preexisting data.
- Generative AI has impressive capabilities and a wide range of possible implementations.
 - Blog entries, code, poetry, FAQ responses, sentiment analysis, artwork, and even films are just some of the **textual and visual outputs** of generative AI models.
- Potential benefits of Generative AI adoption for enterprises
 - Increased productivity
 - Improved and personalized customer experience
 - Boosted creativity
 - Boosted scalability



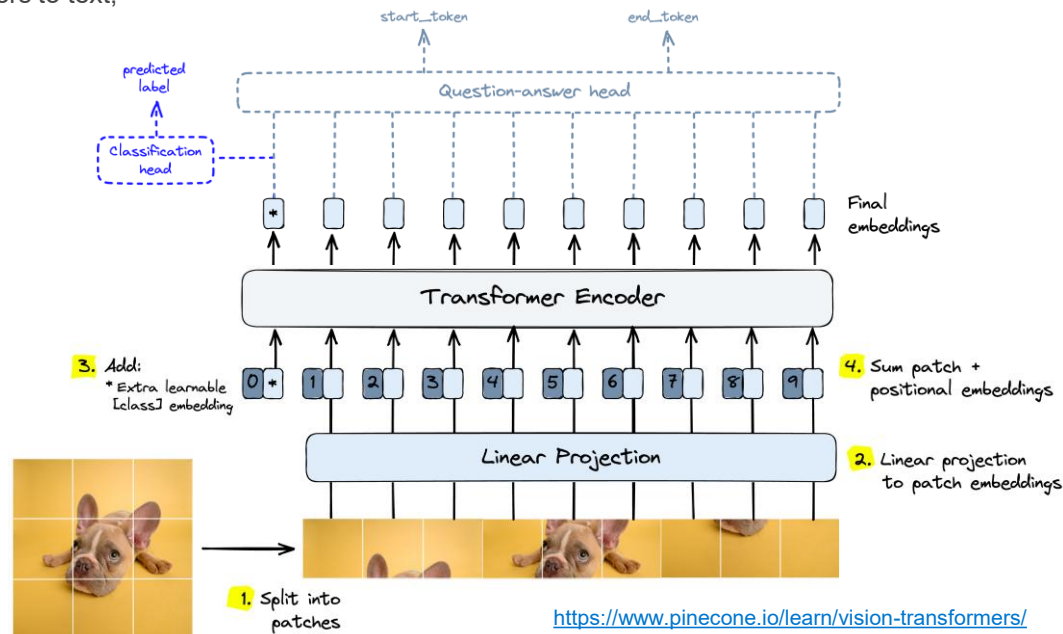
Transformer in Vision

Vision Transformer (ViT)

- Transformers lack the **inductive biases of CNNs**, such as **translation invariance** and a **locally restricted receptive field**. Invariance means that you can recognize an entity (i.e. object) in an image, even when its appearance or position varies. Translation in computer vision implies that each image pixel has been moved by a fixed amount in a particular direction.
- On the other hand, **the transformer is by design permutation invariant**. The ViT model represents an input image as a series of image patches, like the series of word embeddings used when using transformers to text, and directly predicts class labels for the image.
- Multi-head Attention Network is responsible for generating attention maps from the given embedded visual tokens. These **attention maps** help the network focus on **the most critical regions in the image**, such as object(s).

- Split an image into patches
- Flatten the patches
- Produce lower-dimensional linear embeddings from the flattened patches
- Add positional embeddings
- Feed the sequence as an input to a transformer encoder
- Pretrain the model with image labels (fully supervised on a huge dataset)
- Finetune on the downstream dataset for image classification

<https://www.v7labs.com/blog/vision-transformer-guide>

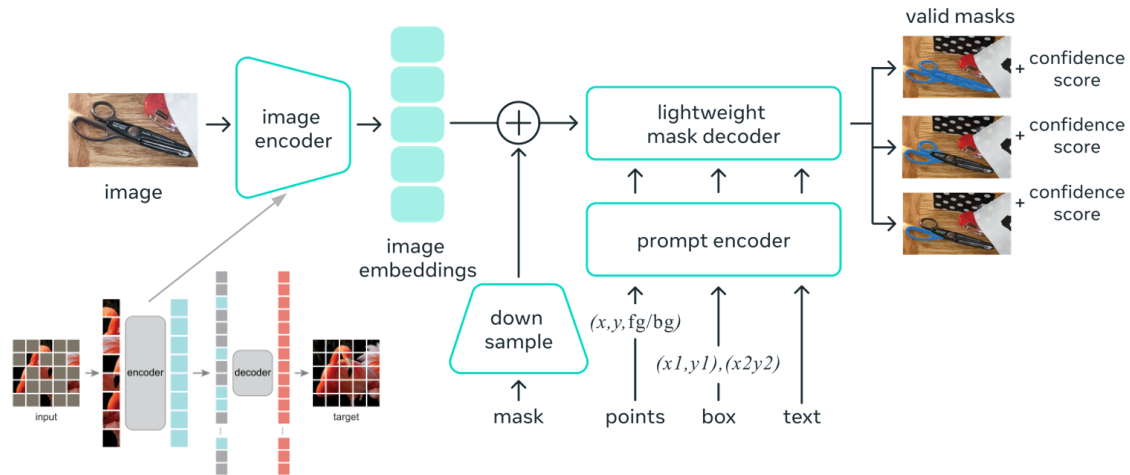


<https://www.pinecone.io/learn/vision-transformers/>

Segment Anything Model (SAM)

- **Segmentation** — identifying which image pixels belong to an object — is a core task in computer vision and is used in a broad array of applications.
- SAM has learned a general notion of what objects are, and it can generate **masks for any object in any image or any video**, even including objects and image types that it had not encountered during training.
- An **image encoder** produces a one-time embedding for the image, while a **lightweight encoder** converts any prompt into an embedding vector in real time. These two information sources are then combined in a **lightweight decoder** that predicts segmentation masks.
- After the image embedding is computed, SAM can produce a segment in just 50 milliseconds given any prompt in a web browser.

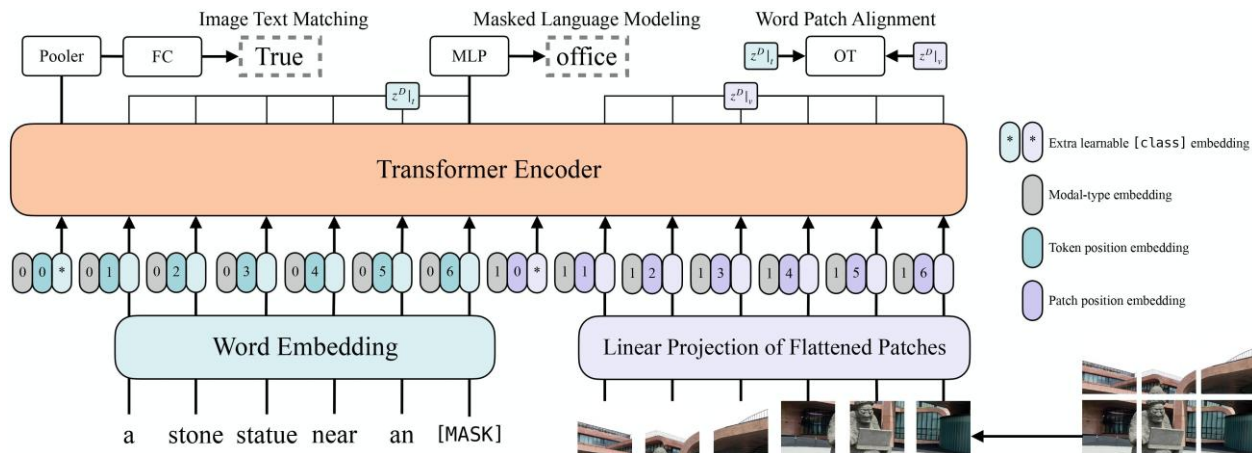
Universal segmentation model



Bridging Vision and Language

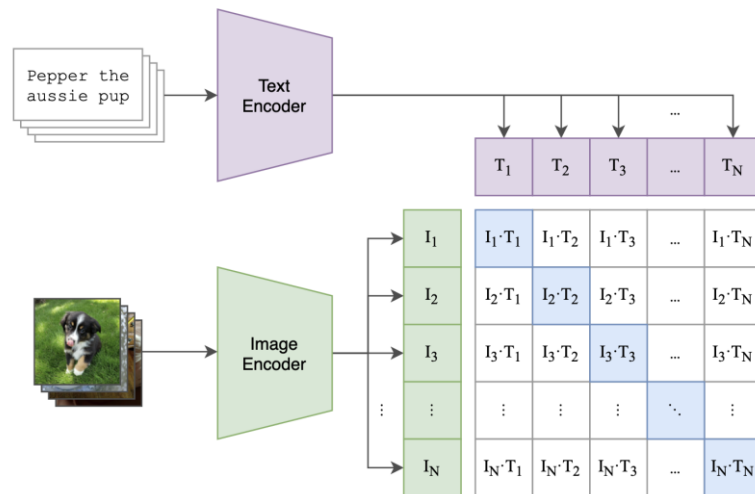
ViLT: Vision-and-Language Transformer

- ViLT is a minimal [vision-and-language pre-training transformer](https://arxiv.org/abs/2102.03334) model where processing of visual inputs is simplified to just the same convolution-free manner that text inputs are processed.
- The model-specific components of ViLT require less computation than the transformer component for multimodal interactions.
- The model is pre-trained on the following objectives: image text matching, masked language modeling, and word patch alignment.
- Vision-and-Language Pre-training has improved performance on various joint vision-and-language downstream tasks.



CLIP: Contrastive Language Image Pretraining

- CLIP is **zero-shot** classifier with widespread implications.
 - CLIP broke through the traditional method of using a pre-trained model like ResNet, which involved collecting huge custom datasets of labelled images.
- CLIP attempts to form **relationships between images and text**, by learning associations from very large amounts of text/image pairs.
 - CLIP was trained on **400 million image/text pairings**, mostly obtained from minimally-curated web-scraped data, i.e., images that are drawn from public internet resources, and which have associated text, either in the form of metadata, such as alt-tags, or more expressly, which have been explicitly captioned.
- Besides being a multimodal (text + image) system, CLIP's utility mainly lies in the sheer breadth of the data that it was trained on.
 - This means that CLIP is capable of performing useful operations on **out-of-distribution (OOD) data**, i.e., images / concepts that it has never been exposed to.

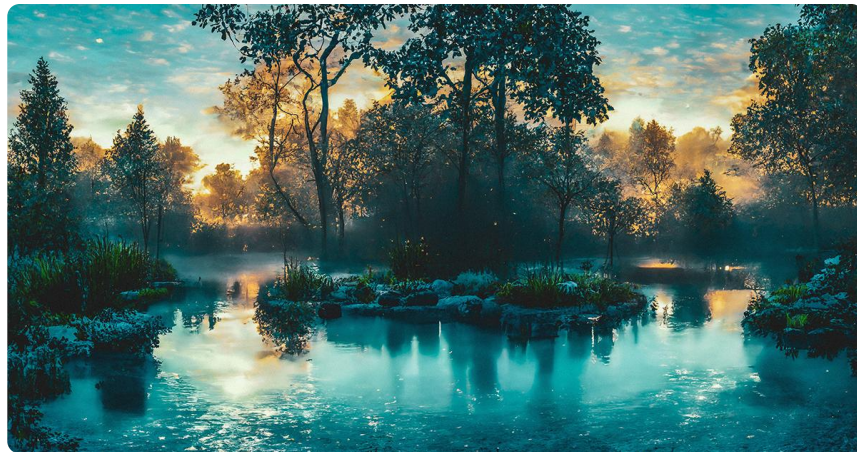


While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples.

Text-to-Image / Video Generator

The Rise of Text-to-Image Generators

- **Diffusion Models** are a conceptually **simple and elegant approach** to the problem of generating data. Current research shows that diffusion models have emerged as a promising generative modeling framework, pushing the state-of-the-art image and video generation tasks.
- Many of today's text-to-image generation models focus on learning to **iteratively generate images based on continual linguistic input**. They are now slowly transforming from generating dreamlike images to producing realistic portraits.
- There is no doubt that quickly advancing text-to-image generator technology is paving the way for unprecedented opportunities for **instant editing and generated creative output**.
- There are also many challenges ahead, ranging from questions about ethics and bias to issues around copyright and ownership.

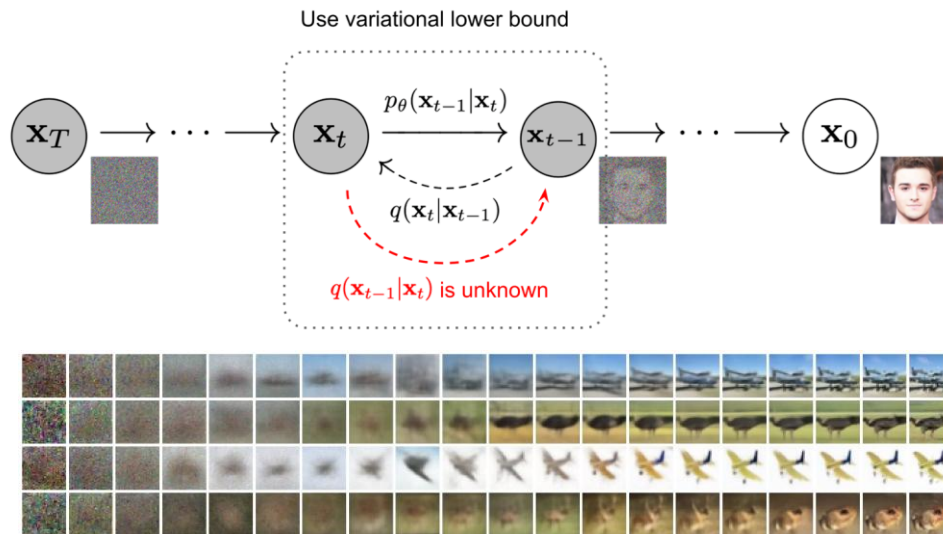


<https://amplify.nabshow.com/articles/ic-ai-image-tools-good-for-creator-economy/>

<https://venturebeat.com/ai/accelerating-ai-for-growth-the-key-role-of-infrastructure/>

Diffusion Models

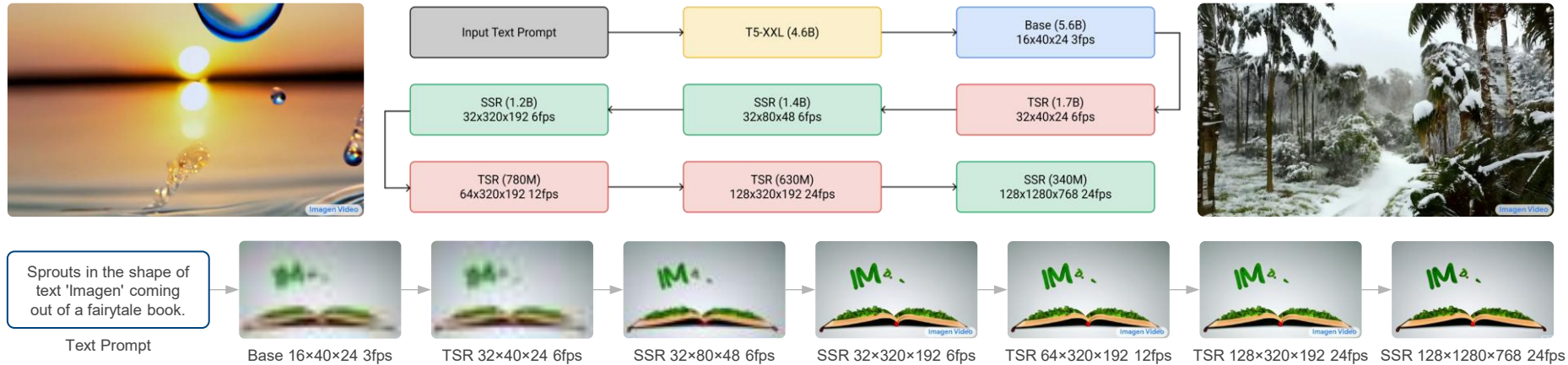
- Diffusion Models are **generative models** which have been gaining significant popularity in the past several years.
 - Inspired by non-equilibrium thermodynamics, Diffusion Models currently produce State-of-the-Art image quality.
- Diffusion Models are used to generate data similar to the data on which they are trained.
 - Fundamentally, Diffusion Models work by destroying training data through the **successive addition of Gaussian noise**, and then learning to **recover the data by reversing this noising process**.
 - After training, we can use the Diffusion Model to generate data by simply passing randomly sampled noise through the learned denoising process.
 - A Diffusion Model consists of a forward process (or diffusion process), in which a datum (generally an image) is progressively noised, and a reverse process (or reverse diffusion process), in which noise is transformed back into a sample from the target distribution.



GLIDE: Photorealistic Image Generation

- OpenAI's GLIDE is one of the many exciting projects working towards generating and editing photorealistic images using text-guided diffusion models.
 - Interpreting the meaning of a string and translating that message to an image is a task that even humans can struggle with. This is largely why text to visual models like CLIP tend to pop up so frequently in research on this subject.
 - The most popular and common use case for GLIDE will undoubtedly be its potential for image synthesis, its potential for one shot image generation is near limitless.
 - The more interesting application of GLIDE is its automated photo inpainting. GLIDE is capable of taking in an existing image as input, process it with the text prompt in mind for areas that need to be edited, and can actively make additions to an image to those regions with ease. Applications leveraging abilities such as this are potentially key for creating code-less methods for image editing.
- The GLIDE architecture can be boiled down to three parts
 - An Ablated Diffusion Model (ADM) trained to generate a 64 x 64 image
 - A text model (transformer) that influences that image generation through a text prompt
 - An upsampling model that takes our small 64 x 64 images to a more interpretable 256 x 256 pixels.
 - The first two components interact with one another to guide the image generation process to accurately reflect the text prompt, and the latter is necessary to make the images we produce more easy to interpret.

Imagen Video: Text-to-Video Generator



- **Google's Imagen Video** is a text-to-video generative AI model capable of producing high-definition videos from a text prompt. The text-conditioned **video diffusion model** can generate videos up to a resolution of 1280x768 at 24 frames per second.
- Imagen Video consists of a **text encoder** (frozen T5-XXL), a **base video diffusion model**, and **interleaved spatial and temporal super-resolution diffusion models**.
- Imagen Video builds on Google's own Imagen text-to-image

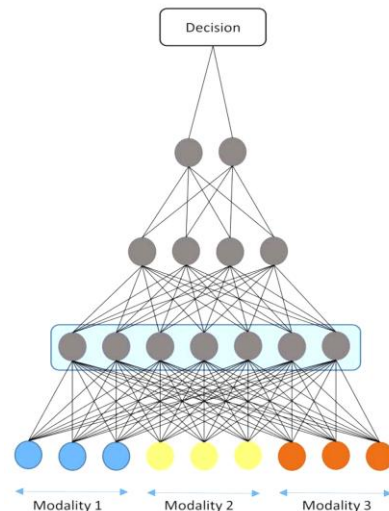
synthesis models. In fact, one of the biggest contributions of Imagen Video was to many of the architecture principles of Imagen are **transferrable to video generation settings**. Both techniques are based on diffusion architectures which have become the foundation for modern research image and video generation.

- The cornerstone of Imagen Video is a **cascade architecture of video diffusion models**. The greatest benefit of cascaded diffusion methods is their ability to models high-dimensional problems by using a combination of relatively simple sub-models.

Multimodal Learning with Transformer

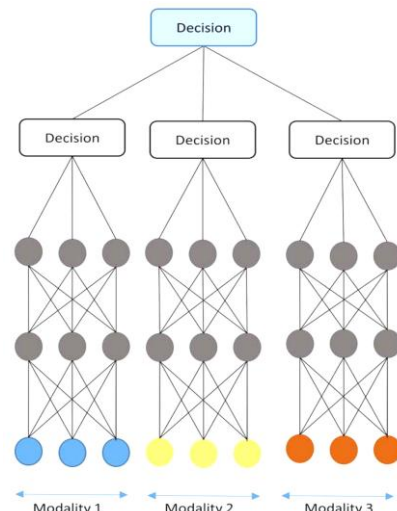
Multimodal Learning with Transformers

- The Transformers are officially the common architecture we needed for multimodal learning.
 - The breakthrough of Transformer-based pre-trained models in NLP has inspired academic interest in the **convergence of several modalities**, such as video and text or image and text.
- Large pre-trained models have been proven as the most effective framework for **joint modeling of multi-modality data**.
 - Multimodal pre-trained models based on Transformers can **learn semantic correspondence between different modalities** by pre-training on large amounts of unlabeled data and then fine-tuning on small amounts of labeled data.
 - They can manipulate text, images, video, audio and any type of data that can be turned into **tokens**.



Early Fusion

In 'early fusion', the original input data are **concatenated**, and the resulting vector is treated like unimodal input. Early fusion has its advantages in its **simplicity** because no design choices about how to extract marginal representations have to be made.



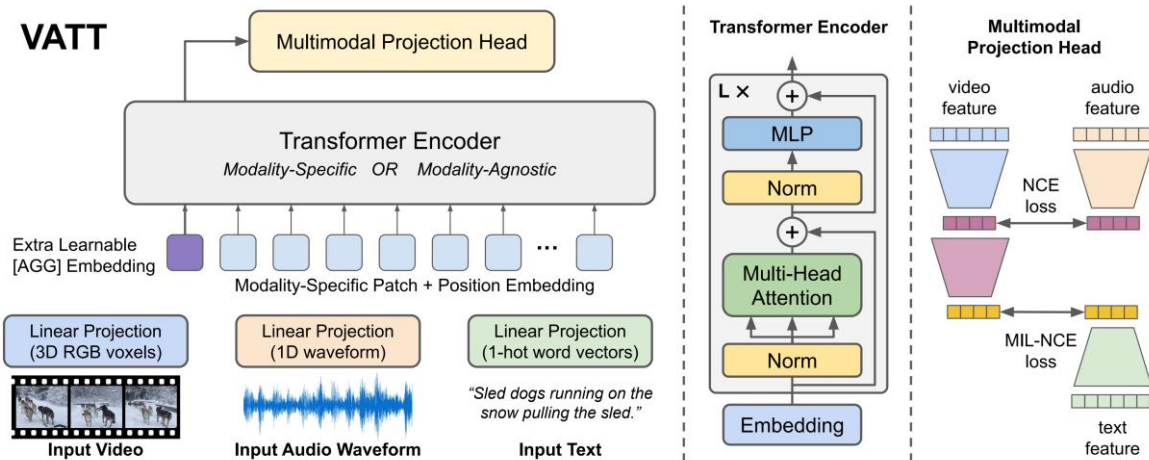
Late Fusion

In 'late fusion', instead of combining the original data or learned features, **decisions of separate unimodal sub-models are combined** into a final decision. This allows learning good marginal representations since each model can be adapted to the specific modality.

<https://academic.oup.com/bib/article/23/2/bbab569/6516346>

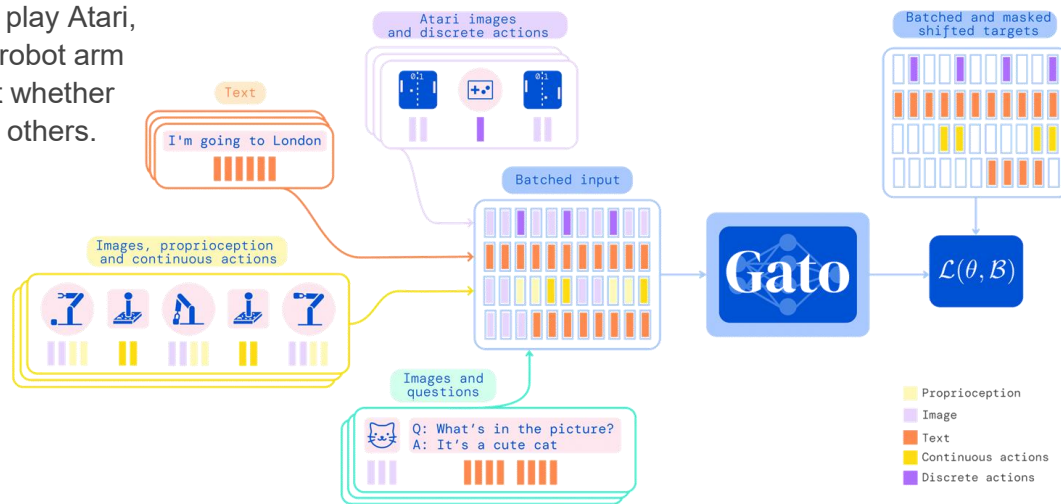
VATT: Video, Audio-Text Transformer

- One of the most important applications of Transformers in the field of Multimodal Learning is certainly VATT.
- This study seeks to exploit the ability of Transformers to **handle different types of data** to create **a single model** that can learn simultaneously from **video, audio and text**.
- The proposed architecture is composed of a single Transformer Encoder on which **three distinct forward calls** are made. One call for each type of input data is always transformed into a sequence of tokens.
- The transformer takes these sequences as input and returns three distinct sets of features.
- Then the features are given in input to a **Contrastive estimation block** that calculates a single loss and performs the backward.



GATO: A Generalist Agent

- A large volume of data scientists all over the world suggests that GATO is the world's first AGI.
- GATO, a **multi-modal, multi-task, multi-embodiment generalist agent**, represents one of the most impressive achievements in this field today.
 - Inspired by progress in large-scale language modelling, a similar approach is applied towards building a **single generalist agent beyond the realm of text outputs**.
 - The same network with the same weights can play Atari, caption images, chat, stack blocks with a real robot arm and much more, deciding based on its context whether to output text, joint torques, button presses, or others.
- Internally, once again, there is a Transformer that takes in input data of different types transformed into a sequence of tokens.
- Thanks to this unification of inputs and to the Transformer architecture, the model will be able to acquire information from even very different sources, achieving an unprecedented level of generalization.
- During the training phase of Gato, data from different tasks and modalities are serialized into a flat sequence of tokens, batched, and processed by a transformer neural network similar to a large language model. The loss is masked so that Gato only predicts action and text targets.



- Mobile Manipulation**

Human: Bring me the rice chips from the drawer. Robot: 1. Go to the drawers, 2. Open top drawer. I see ****. 3. Pick the green rice chip bag from the drawer and place it on the counter.

Visual Q&A, Captioning ...

Given ****: Q: What's in the image? Answer in emojis.
A: 🍌🥑🍎🍓🍇🍓🍌

PaLM-E: An Embodied Multimodal Language Model

Given **<emb>** ... ****: Q: How to grasp blue block? A: First, grasp yellow block

The diagram illustrates the internal processing of PaLM-E, showing a sequence of tokens (orange boxes) being processed by a Vision Transformer (ViT, blue box) and a Large Language Model (PaLM, orange bar). The output is a Control signal (purple box) which generates a sequence of actions (orange boxes) like "A: First, grasp yellow block and ...".

Task and Motion Planning

Given **<emb>**: Q: How to grasp blue block?
A: First grasp yellow block and place it on the table, then grasp the blue block.

Tabletop Manipulation

Given ****: Task: Sort colors into corners.
Step 1. Push the green star to the bottom left.
Step 2. Push the green circle to the green star.

Language Only Tasks

Here is a Haiku about embodied language models:
Embodied language models are the future of natural language

Q: Miami Beach borders which ocean? A: Atlantic.
Q: What is 372 x 187? A: 6696.
Language models trained on robot sensor data can be used to guide a robot's actions.

- According to Google, when given a high-level command, such as "bring me the rice chips from the drawer," PaLM-E can generate **a plan of action for a mobile robot platform** with an arm (developed by Google Robotics) and execute the actions by itself.
- PaLM-E does this by analyzing data from the robot's camera without needing a pre-processed scene representation.

<https://ai.googleblog.com/2023/03/palm-e-embodied-multimodal-language.html>

